

Design Proposal: Mach-Zehnder Interferometer

David Zambrano dzambrano2594@floridapoly.edu UBCx: DavidZambano28

Department of Electrical & Computer Engineering, Florida Polytechnic University, Lakeland, Florida, United States

Abstract—This document is a design proposal for a Mach-Zehnder Interferometer to be fabricated on a 220nm Silicon-on-Insulator process. Included is fundamental theory, analytic results obtained through commercial software, and a justification for the design decisions made.

Index Terms—Interferometry, Mach-Zehnder Interferometer, Silicon Photonics, Silicon-on-insulator.

I. INTRODUCTION

WITHIN the field of integrated photonics, the Mach-Zehnder Interferometer is a fundamental building block. It has many applications, such as optical switches [1] or biosensing [2] given its operation principle. Geometrically, it is a very simple device, generally consisting of two Y-branch waveguides connected with waveguides. In this proposal, we will explore the operation of an imbalanced Mach-Zehnder Interferometer, which has the connecting waveguides that share a path length difference, contributing to a phase shift at the output.

II. THEORY

As mentioned, the Mach-Zehnder interferometer generally consists of two Y-branch waveguides. These structures have unique physics that classify the total geometry as a Mach-Zehnder interferometer. Figure 1 shows the structure, with the left port being the input port, which branches off into two output ports. The Y-branch functions as a beam splitter, which, in the ideal case, has the light split equally between the two branches in which 50% of the light intensity of the fundamental mode is found in either output port.

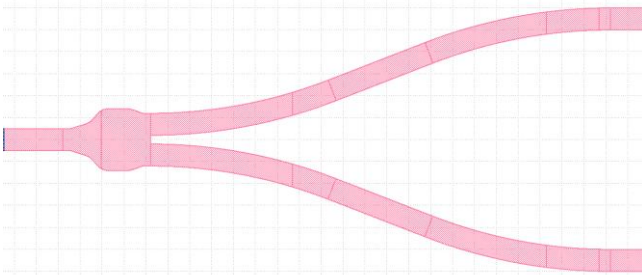


Fig. 1. Y-Branch Waveguide (KLayout E-Beam PDK).

This is expressed as

$$I_1 = I_2 = \frac{I_i}{2} \quad (1)$$

Where I_1 and I_2 are the output intensities found in output 1 and 2 respectively. In terms of the electric field, a relationship $I_i \propto |E_i|^2$ exists in which we can define the electric field at output 1 and 2 as

$$E_1 = E_2 = \frac{E_i}{\sqrt{2}}. \quad (2)$$

In the case where two waveguides connect two Y-branches we arrive at the Mach-Zehnder Interferometer as shown in Figure 2.

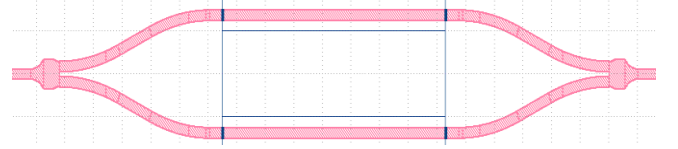


Fig. 2. Y-Branch formed Mach-Zehnder Interferometer (Klayout E-Beam PDK).

Here, the Y-branch waveguide on the right serves as beam combiner rather than a splitter, as its branches are now inputs combining into one physical output. Here, multiple modes including the fundamental mode and higher order modes (radiation modes) comprise the total output intensity and electric field. For the fundamental mode, the output electric field is described as

$$E_o = \frac{E_1 - E_2}{\sqrt{2}}. \quad (3)$$

And the output intensity as

$$I_o = \frac{1}{2} I_i [1 + \cos(\phi)] \quad (4)$$

Where ϕ is the phase shift caused by the optical path length difference, in which the output intensity I_o switches between 0 and 1 when the phase varies between 0 and 180 degrees. For ϕ , we can write

$$\phi = \beta \Delta L \quad (5)$$

Where ΔL is the path length difference, and β is the propagation constant of the waveguide

$$\beta = \frac{2\pi n}{\lambda} \quad (6)$$

n being the effective index of the waveguide, and λ is the wavelength of light. Using this, we can adjust the ratio of light intensity $\frac{I_o}{I_i}$ arriving at the output of the Mach-Zehnder Interferometer.

III. MODELING AND SIMULATION

A. Waveguide

For this design, a simple 500x220nm (WxH) silicon strip waveguide is selected. This geometry will comprise the entire device between the Y-branches and connecting waveguides. The operating wavelength is at about the center of the optical C-Band, 1550nm [3]. This 220nm etch depth is the standard for Multi-Project Wafer runs, which will be performed by Applied Nanotools' NanoSOI service [4]. Using Lumerical MODE, we can extract the mode profiles of this geometry

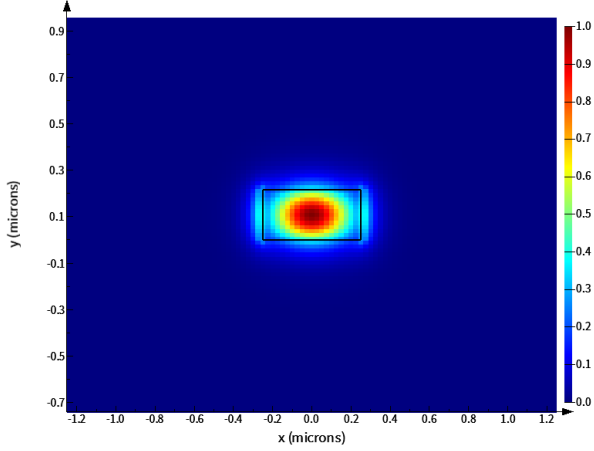


Fig. 3. Electric field intensity of 500x220nm waveguide (TE Mode).

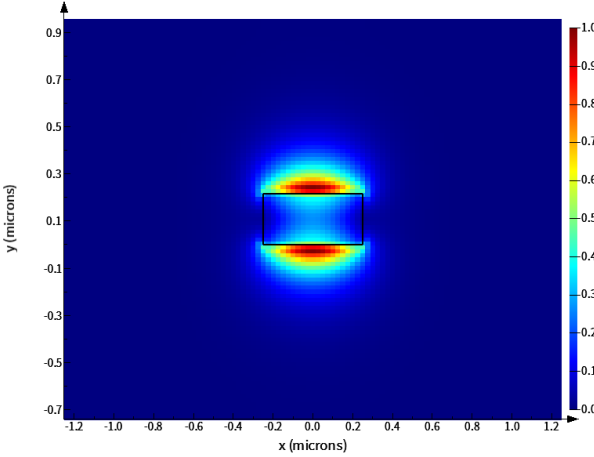


Fig. 4. Electric field intensity of 500x220nm waveguide (TM Mode).

We can see good optical confinement within the Transverse Electric (TE) mode, which will be the polarization of choice for this design. In addition to the mode profiles, we extract the effective and group index of the waveguide in Lumerical MODE.

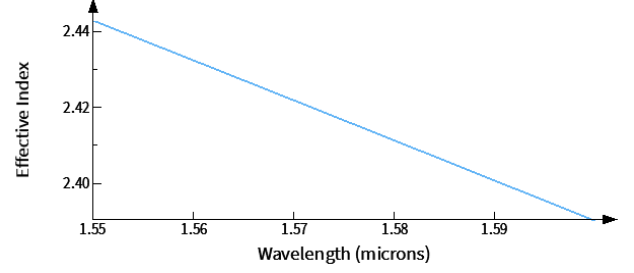


Fig. 5. Effective index vs. wavelength (TE Mode).

We can see how the effective index decreases with wavelength. For the first modes, the effective index at a 1550nm wavelength can be seen in Table 1.

TABLE I
EFFECTIVE INDEX OF SIMULATED WAVEGUIDES AT 1550NM

MODE	EFFECTIVE INDEX
TE0	2.4429
TM0	1.7666
TE1	1.4952
TE2	1.3465

We can also plot the group index to see a different relationship vs. wavelength.

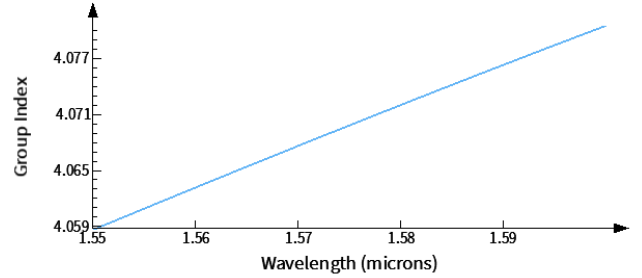


Fig. 6. Group index vs. wavelength (TE Mode).

While the effective index decreases with wavelength due to material dispersion and the relationship found in (6). The group index, however, decreases with wavelength. The group index can be described as

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda \frac{dn_{eff}}{d\lambda} \quad (7)$$

Where a direct relationship with the effective index exists. Since the effective index generally decreases with wavelength, we should expect the group index to have the inverse relationship. For the purposes of verification, the group index will be the primary comparison parameter between the analytic and experimental results. A compact model of the waveguide can be described by

$$n_{eff}(\lambda) = 2.44 - 1.13(\lambda - \lambda_0) - 0.04(\lambda - \lambda_0)^2 \quad (8)$$

For the effective index vs. wavelength.

B. Y-Branch Waveguide

The Y-branch waveguide will have the same 500x220nm geometry as the connecting waveguides to form its paths. We can verify that the structure (Y-Branch in the EBeam PDK in Klayout) functions as a beam splitter. We do this with Lumerical 2.5D variational FDTD solver. By placing a mode source at the input port and a frequency domain monitor at the input port and one of the branch ports, we can see the correct behavior.

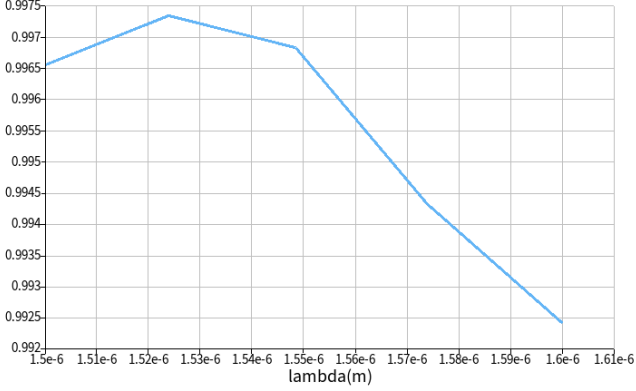


Fig. 7. Transmission vs. wavelength (input port).

Transmission as a ratio of input power to measured power (0 to 1) shows that the input port we have almost 100% of the power passing through minus some expected losses.

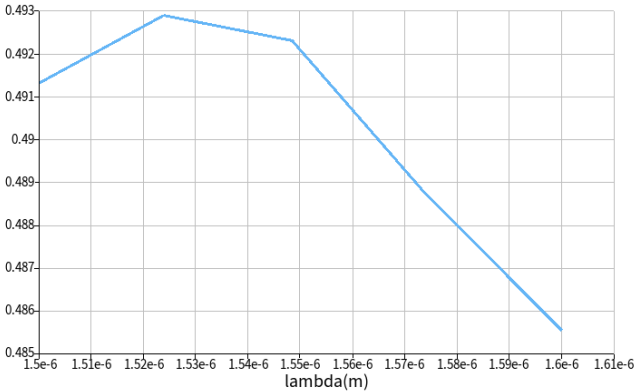


Fig. 8. Transmission vs. wavelength (top branch port).

Then, we can see ~50% of the input power arriving at the top branch port, as expected.

C. Mach-Zehnder Interferometer

With these two devices modeled, we can then use Lumerical INTERCONNECT to model the MZI in full. The transfer function of the MZI is defined as

$$T_{MZI} = \frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\phi)] \quad (9)$$

Obtained from the earlier (4). To model the MZI as accurately as possible, we connect grating couplers (GC) to the Y-branch waveguides to couple light into and out of the circuit. We use S-parameter components derived from the MODE and FDTD simulations for the Y-branch and GCs. We simulate from

1500-1600nm to fully analyze the C-band spectrum of the MZI.

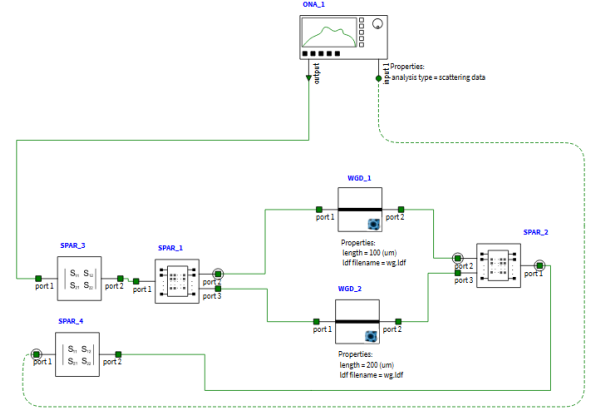


Fig. 9. Lumerical INTERCONNECT simulation in full.

Figure 9 shows the complete INTERCONNECT circuit. Using an optical network analyzer (ONA), we can obtain our desired measurements. Firstly, we must consider the importance of incorporating the GCs into the simulation.

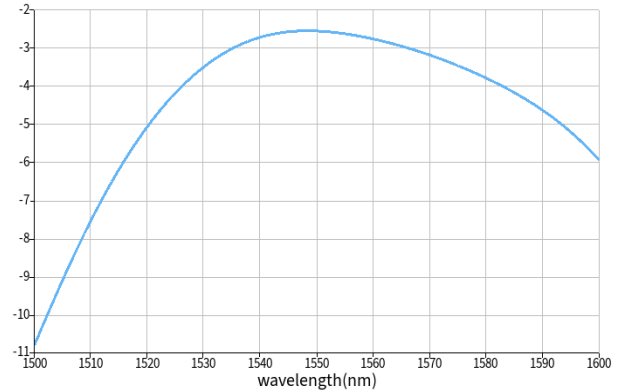


Fig. 10. GC loss vs. wavelength (dB).

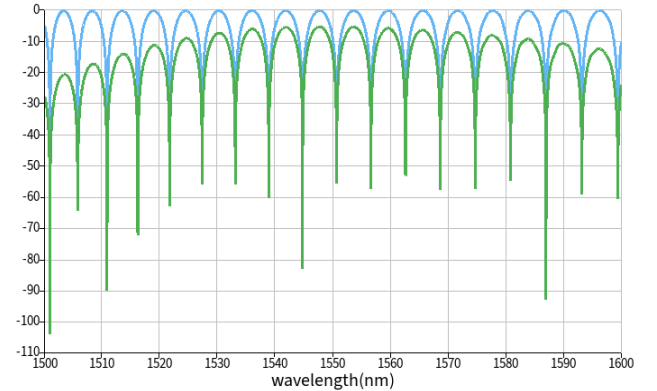


Fig. 11. MZI Transfer function with GC (green) & without (blue).

Alone, we can see about a 2.5dB loss at our 1550nm

wavelength. Compounded between the input and output, however, results in a large difference in the response as seen in Figure 11. With that established, we can then design our MZI around the path length difference ΔL . In particular, the π phase shift provides the most interesting result. To determine the appropriate path length difference, we use (5) and (6) to solve for our ΔL at $\lambda = 1550\text{nm}$ and $n = 2.44$. Doing so gives us $\Delta L = 317.6\text{nm}$.

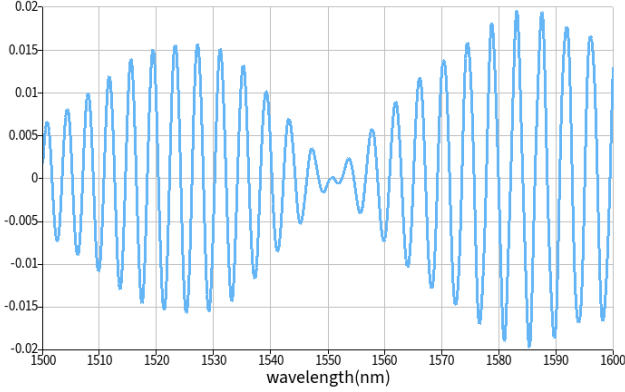


Fig. 12. Transmission (Real) with $\Delta L = 317.6\text{nm}$

We can see how the simulated transmission plot meets our expectations at 1550nm, where the transmission goes to (or near) 0.

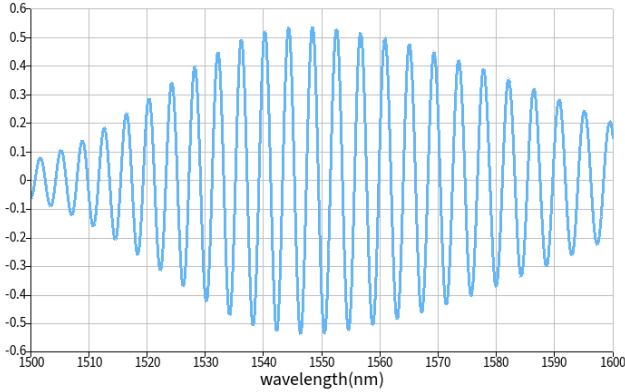


Fig. 13. Transmission (Real) with $\Delta L = 0$

In comparison, the balanced MZI with no path length difference shows constructive interference effects across the entire spectrum. Since a destructive interference π phase shift can only exist at an odd integer multiple of our found ΔL , we can design the path length difference to be say $25.0922\mu\text{m}$ ($m = 29$). The design then will focus on odd integer multiples to achieve these destructive effects, as well as other ΔL values for other output ratios. Lastly, we will consider the Free Spectral Range (FSR) of these devices. Measurement dictates that the FSR be no larger than 50nm. This can be verified in INTERCONNECT, a list of values at chosen ΔL shown in Table 2.

TABLE II
FSR AT MULTIPLE ΔL AT $\sim 1550\text{nm}$

ΔL (μm)	FSR (nm)
25.0922	22.1
40.3381	7.23
73.3709	7.75
100.0512	5.71

It is likely that the exact path length differences will vary or be rounded up in the design software, but these values serve to prove the FSR will be within the acceptable range.

$$FSR(\lambda) = \frac{\lambda^2}{\Delta L n_g(\lambda)} \quad (10)$$

Using these values, we can obtain the group index by solving for $n_g(\lambda)$ in (9). Doing so at $\Delta L = 73.3709$ results in $n_g(\lambda) = 4.225$. The calculation of the group index will be used as the primary comparison between the simulation and experimental results.

IV. CONCLUSION

In this paper, we went over the basic device physics of the Mach-Zehnder Interferometer for the purpose of proposing an unbalanced interferometer design, specifically to highlight destructive interference effects. While the current progress exists only in simulation, experimental results obtained through automated measurement the designs fabricated on a 220nm Silicon-on-Insulator process will eventually validate these findings.

REFERENCES

- [1] Md. S. Islam and M. J. Barsha, "Mach Zehnder Interferometer (MZI) as A Switch for All Optical Network," in *2018 International Conference on Innovation in Engineering and Technology (ICIET)*, Dec. 2018, pp. 1–5. doi: 10.1109/CIET.2018.8660861.
- [2] J. Montoya-Cardona *et al.*, "Performance Analysis of a Mach-Zehnder Interferometer Based on Dual-Core Transversally Chirped Microstructured Optical Fiber for Biosensing applications," in *2018 International Conference on Electromagnetics in Advanced Applications (ICEAA)*, Sep. 2018, pp. 384–387. doi: 10.1109/ICEAA.2018.8520405.
- [3] "BYU Cleanroom - ITU Wavelength Grid: DWDM." Accessed: Oct. 04, 2026. [Online]. Available: <http://www.photonics.byu.edu/FiberStandards.parts/DWDM.phtml>
- [4] "NanoSOI Fabrication Service | Applied Nanotools Inc." Accessed: Oct. 04, 2026. [Online]. Available: <https://www.appliednt.com/nanosoi-fabrication-service/>